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Inequations

Inequations, also known as inequalities, represent relationships between two quantities where they are not necessarily equal. They use symbols such as $>$, $<$, $\geq$, and $\leq$ to indicate whether one quantity is greater than, less than, greater than or equal to, or less than or equal to the other. Inequations can involve linear expressions, indicating a range of possible solutions, or more complex forms like absolute value inequalities, requiring consideration of multiple cases to capture all possible solutions. When solving inequalities, the focus is on isolating the variable while maintaining the relationship denoted by the inequality symbol, with special care taken when multiplying or dividing by negative numbers. Inequations are key in algebra, calculus, and optimization, providing a way to describe conditions where equality does not necessarily hold.

Example 5: Solve $x^2 + 4x - 12 > 0$

To solve the inequality $x^2 + 4x - 12 > 0$, is possible to use the quadratic formula or by factoring the quadratic expression:
To factorize, we find two numbers whose product is -12 and whose sum is 4: the numbers 6 and -2 satisfy these conditions.
Thus, the factorized form of the quadratic is:

$$(x + 6) \cdot (x - 2)$$

Setting the factorized form to zero, we get the roots:
1. $x + 6 = 0 \Rightarrow x = -6$,
2. $x - 2 = 0 \Rightarrow x = 2$.

Given the roots at $x = -6$ and $x = 2$, we examine the intervals between and beyond these points to determine where the inequality holds true.
When x $<$ - 6, the expression is positive.
When -6 $<$ x $<$ 2, the expression is negative.
When x $>$ 2, the expression is positive.

$-\infty$ ...	-6	...	2	... $+\infty$
> 0	$= 0$	< 0	$= 0$	> 0
$\uparrow$				$\uparrow$

Thus, the intervals where $x^2 + 4x - 12 > 0$ are: $x \in (-\infty, -6[\cup]2, +\infty)$
These intervals represent where the quadratic expression is positive, indicating where the inequality holds true.

Example 6: Solve $(x-\frac{7}{2})^2 \cdot (3x+3) \cdot (2x-1) > 0$

To solve the inequality $(x-\frac{7}{2})^2 \cdot (3x+3) \cdot (2x-1) > 0$, you need to find the critical points where the expression equals zero, then use these points to determine the sign of the inequality in each interval.
The critical points occur when the expression is zero. Set each factor to zero and solve for x:
1. $x-\frac{7}{2}=0 \Rightarrow x=\frac{7}{2}$
2. $3x+3=0 \Rightarrow x=-1$
3. $2x-1=0 \Rightarrow x=\frac{1}{2}$

Thus, the critical points are: $x=\frac{7}{2}$, $x=-1$, $x=\frac{1}{2}$.
Now, consider the intervals between and beyond the critical points to determine where the expression is greater than zero.
Given the inequality has roots at $x=-1$, $x=\frac{1}{2}$, and $x=\frac{7}{2}$, let's examine the intervals:
When x < -1, the expression is positive.
When -1 < x < $\frac{1}{2}$, the expression is negative.
When $\frac{1}{2}$ < x < $\frac{7}{2}$, the expression is positive.
When x > $\frac{7}{2}$, the expression is positive

$-\infty$ ...	-1	...	$\frac{1}{2}$	...	$\frac{7}{2}$	... $+\infty$
>0	$=0$	<0	$=0$	>0	$=0$	>0
$\uparrow$				$\uparrow$		$\uparrow$

Thus, the intervals where $(x-\frac{7}{2})^2 \cdot (3x+3) \cdot (2x-1) > 0$ are:
$x \in (-\infty,-1[\cup]\frac{1}{2},\frac{7}{2}[\cup]\frac{7}{2},+\infty)$
These intervals represent the regions where the given inequality holds true.